

Appendix D German Credit

Key references include Häußler (1979); UCI Machine Learning Repository (1994); Grömping (2019); UCI Machine Learning Repository (2019).

D.1 Datasheet

D.1.1 Motivation

- **For what purpose was the dataset created?**
This dataset was created to study the problem of automated credit decisions at a regional Bank in southern Germany.
- **Who created the dataset and on behalf of which entity?**
The dataset was created at a regional Bank of southern Germany (most likely Hypo Bank) and first used by Walter Häußler in the late 1970s as part of his PhD thesis. Hans Hofmann, affiliated with Universität Hamburg at the time, is credited as dataset source (UCI Machine Learning Repository, 1994). Presumably, he donated the dataset to the European Statlog project and a representative of Strathclyde University donated it to UCI (Grömping, 2019).
- **Who funded the creation of the dataset?**
The first known work using the dataset describes it as originating from a regional Bank of southern Germany (Häußler, 1979). Given the affiliation of the author is Hypo Bank, which fit the description at the time, we assume the dataset was collected, curated and funded at Hypo Bank.

D.1.2 Composition

- **What do the instances that comprise the dataset represent?**
Instances represent Hypo bank **loan recipients** from 1973–1975.
- **How many instances are there in total?**
The dataset consists of **1,000** instances.
- **Does the dataset contain all possible instances or is it a sample of instances from a larger set?**
In principle this is a **convenience sample**, consisting of people who were deemed creditworthy by a bank clerk. A representative sample stemming from indiscriminate credit grants would not have been viable (Häußler, 1979). However, if the envisioned application was *post-screening* credit decisions, the influence of this selection bias would be reduced. Finally loan recipients associated with delayed payment or loan default (“bad credit”) are oversampled (30%).
- **What data does each instance consist of?**
For each instance, 13 categorical and 7 quantitative variables are provided, summarizing their financial situation, credit history, and personal situation, including housing, number of liable people, and a mixed variable encoding marital status and sex. A more thorough description is deferred to Tables 27-29.
- **Is there a label or target associated with each instance?**
Yes. A binary label encodes whether loan recipients punctually payed each installment (“good credit”) or not (“bad credit”). The latter label includes a range of situations from delayed payment up to loan default.
- **Is any information missing from individual instances?**
No. No cell is missing, however the variable “property” has a level jointly encoding the conditions “no property” and “unknown”. A similar joint encoding exists for “savings”, so some values may actually be deemed missing for these variables.
- **Are relationships between individual instances made explicit?**
No. There are no known relationships between instances.

- **Are there recommended data splits?**
No.
- **Are there any errors, sources of noise, or redundancies in the dataset?**
Yes. The dataset documentation is filled with errors, so that several levels of categorical variables do not correspond to what they should according to the official documentation from UCI Machine Learning Repository (1994). This is not necessarily an issue if one is purely interested in the evaluation of a method. For example, according to the official documentation, a majority of loan recipients are foreign workers, while in reality this should appear rather strange and indeed is not true (Grömping, 2019). Computationally, this will make no difference, as the input to a machine learning method will remain the same. However if one is interested to the context surrounding the data, as should be the case with fairness research, the wrong encoding poses several problems. The most significant problem is the impression that one can retrieve people’s sex from the joint sex-marital-status encoding, which is simply false as a single level corresponds to both single males and divorced/separated/married females (Grömping, 2019). Despite this information being available since 2019, the fairness community does not seem to have taken notice. Several experiments of algorithmic fairness on this dataset consider the protected attribute “sex” (sometimes even called “gender”). These experiments are part of work recently published in the most reputable venues for fairness research (Appendix A.73). More mistakes in the documentation of eight variables and the relative errata are outlined in Grömping (2019). A clean version of the dataset is available at UCI Machine Learning Repository (2019).
- **Is the dataset self-contained, or does it link to or otherwise rely on external resources?**
The dataset is **self-contained**.
- **Does the dataset contain data that might be considered confidential?**
Yes. The dataset summarizes customers’ financial and personal situation, including past credit history.
- **Does the dataset contain data that, if viewed directly, might be offensive, insulting, threatening, or might otherwise cause anxiety?**
No.
- **Does the dataset identify any subpopulations?**
Yes. The dataset identifies subpopulation by age and sex. Sex is jointly encoded with marital status and cannot be retrieved, contrary to documentation accompanying the dataset (UCI Machine Learning Repository, 1994). A summary based on amended documentation (Grömping, 2019) is presented in Table 24.

Demographic Characteristic	Values
Percentage of people under-19 years old	0.20%
Percentage of people between 20-29 years old	36.70%
Percentage of people between 30-39 years old	33.20%
Percentage of people between 40-49 years old	17.60%
Percentage of people between 50-59 years old	7.20%
Percentage of people between 60-69 years old	4.40%
Percentage of people over-70 years old	0.70%
Percentage of people who are male : divorced/separated	5.00%
Percentage of people who are female : non-single or male : single	31.00%
Percentage of people who are male : married/widowed	54.80%
Percentage of people who are female : single	9.20%

Table 24: Demographic characteristics of the German credit dataset.

- **Is it possible to identify individuals, either directly or indirectly, from the dataset?**
Likely no, especially given the fact that these records date back to almost 50 years ago. Also, important variables for re-identification, such as ZIP code and date of birth are missing and many other variables are bucketed.
- **Does the dataset contain data that might be considered sensitive in any way?**
Yes. For each instance, the dataset encodes sex, marital status and financial situation.

D.1.3 Collection process

- **How was the data associated with each instance acquired?**
The data was collected by Hypo bank clerks. Some variables were observable (e.g. credit history with the bank), other variables were reported by subjects (e.g. loan purpose).
- **What mechanisms or procedures were used to collect the data?**
Unknown.
- **If the dataset is a sample from a larger set, what was the sampling strategy?**
The so-called “bad credits” are heavily oversampled to make the classification problem more balanced. A natural selection bias is present in the data, as it only consists of applicants who were deemed creditworthy and were thus granted a loan.
- **Who was involved in the data collection process and how were they compensated?**
The data was likely collected by Hypo bank clerks. Walter Häußler was likely involved in sample selection.
- **Over what timeframe was the data collected?**
The dataset covers loans granted in the period **1973–1975**. Its first publicly-known use dates back to 1979 (Häußler, 1979). It became publicly available in November 1994 (UCI Machine Learning Repository, 1994).
- **Were any ethical review processes conducted?**
Unknown.
- **Was the data collected from the individuals in question directly, or obtained via third parties or other sources?**
Likely both. Some variables were necessarily collected from loan applicants (e.g. loan purpose), while other variables were likely available from bank records (e.g. credit history with the bank).
- **Were the individuals in question notified about the data collection?**
Individuals provided some of this data as part of a loan application. Collection and notification practices for variables like credit history are unclear.
- **Did the individuals in question consent to the collection and use of their data?**
Likely yes, for the purposes of the immediate credit decision. However it seems implausible they agreed to their data becoming publicly available in an anonymized fashion.
- **If consent was obtained, were the consenting individuals provided with a mechanism to revoke their consent in the future or for certain uses?**
Likely no.
- **Has an analysis of the potential impact of the dataset and its use on data subjects been conducted?**
Unknown.

D.1.4 Preprocessing/cleaning/labelling

- **Was any preprocessing/cleaning/labeling of the data done?**
Yes. Some instances were discarded. Remaining instances were associated with a binary label according to compliance with the contract. Bucketing took place on several variables, including balance on checking and savings account (A1, A6) and duration of current employment (A7). Sex and marital status were jointly coded (A9).

- **Was the “raw” data saved in addition to the preprocessed/cleaned/labeled data?**
Unknown.
- **Is the software used to preprocess/clean/label the instances available?**
Likely no.

D.1.5 Uses

- **For what tasks has the dataset been used?**
The dataset was originally used to study the problem of automated credit scoring (Häußler, 1979). Similarly to the Adult dataset, since becoming publicly available it has been used as a benchmark in various machine learning fields.
- **Is there a repository that links to any or all papers or systems that use the dataset?**
Yes. A selection of early works (pre-2005) using this dataset can be found in UCI Machine Learning Repository (1994). A more recent list is available under the beta version of the UCI ML Repository.³¹ See Appendix A.73 for a (non-exhaustive) list of algorithmic fairness works using this resource.
- **What (other) tasks could the dataset be used for?**
The German Credit could be used in fields that concentrate on socially relevant goals and require socially relevant data, such as privacy and explainability. The task at hand is always credit scoring.
- **Is there anything about the composition of the dataset or the way it was collected and preprocessed/cleaned/labeled that might impact future uses?**
Contrary to documentation accompanying the dataset (UCI Machine Learning Repository, 1994), the sex of loan recipients cannot be reliably retrieved. Works of algorithmic fairness should not use this feature.
- **Are there tasks for which the dataset should not be used?**
In its most common version (UCI Machine Learning Repository, 1994) the German Credit dataset should not be used in works of explainability/interpretability as the incorrect documentation would result in counter-intuitive explanations. The 2019 version (UCI Machine Learning Repository, 2019) associated with the erratum (Grömping, 2019) is recommended.

D.1.6 Distribution

- **Is the dataset distributed to third parties outside of the entity on behalf of which the dataset was created?**
Yes. The dataset is publicly available (UCI Machine Learning Repository, 1994)
- **How is the dataset distributed?**
The dataset is available as a **csv file**.
- **When was the dataset distributed?**
The dataset was released to the UCI ML Repository in **November 1994**.
- **Is the dataset distributed under a copyright or other intellectual property (IP) license, and/or under applicable terms of use (ToU)?**
Yes. The UCI ML repository has a citation policy.
- **Have any third parties imposed IP-based or other restrictions on the data associated with the instances?**
Likely no. We are unaware of any IP-based restrictions.
- **Do any export controls or other regulatory restrictions apply to the dataset or to individual instances?**
Unknown.

³¹ <https://archive-beta.ics.uci.edu/ml/datasets/144>

D.1.7 Maintenance

- **Who is supporting/hosting/maintaining the dataset?**
The dataset is hosted and maintained by the **UCI Machine Learning Repository** (UCI Machine Learning Repository, 1994). A clean and well-documented version of the same dataset donated by Ulrike Gromping (UCI Machine Learning Repository, 2019) is also available on the same repository.
- **How can the owner/curator/manager of the dataset be contacted?**
The dataset donor, Hans Hofmann retired in 2008. Comments and inquiries for UCI may be sent to ml-repository@ics.uci.edu.
- **Is there an erratum?**
Yes. A clean data release (UCI Machine Learning Repository, 2019) and accompanying report (Grömping, 2019) are available online.
- **Will the dataset be updated?**
Likely no. The recently released South German Credit Data Set (UCI Machine Learning Repository, 2019) may be considered an update.
- **If the dataset relates to people, are there applicable limits on the retention of the data associated with the instances?**
Unknown.
- **Will older versions of the dataset continue to be supported/hosted/maintained?**
Unless otherwise indicated, both the new (UCI Machine Learning Repository, 2019) and the old version (UCI Machine Learning Repository, 1994) of the German Credit dataset will remain hosted on the UCI ML Repository in its current version.
- **If others want to extend/augment/build on/contribute to the dataset, is there a mechanism for them to do so?**
Unknown.

D.2 Data Nutrition Label

For the sake of correctness, we report redacted information based on the new South German Credit Data Set (UCI Machine Learning Repository, 2019) and accompanying documentation (Grömping, 2019).

D.2.1 Metadata

METADATA	
FileNames	SouthGermanCredit
Format	.asc
Url	https://archive.ics.uci.edu/ml/datasets/South+German+Credit
Domain	Economics
Keywords	credit scoring, Germany, loan, classification
Type	Tabular
Rows	1000
Columns	21
% missing cells	0%
Rows with missing cells	0%
License	UCI Repository citation policy
Released	November 2019
Range	1973-1975
Description	This dataset encodes socio-economical features of loan recipients from a bank in southern Germany, along with binary variable encoding whether they punctually payed every installment, which is he target of a classification task.

Table 25: Metadata of South German Credit dataset.

D.2.2 Provenance

PROVENANCE	
Source	
Name	Walter Häußler
Url	https://archive.ics.uci.edu/ml/datasets/Statlog+%28German+Credit+Data%29
email	//
Authors	
Names	Ulrike Grömping
Url	https://archive.ics.uci.edu/ml/datasets/South+German+Credit
email	groemping@bht-berlin.de

Table 26: Provenance of South German Credit dataset

D.2.3 Variables

VARIABLES	
status	Checking account balance (in Deutsche Mark) 1 (no checking account) 2 (< 0 DM) 3 ($0 \leq \dots < 200$ DM) 4 (≥ 200 DM)
duration	Credit duration (in months)
credit_history	Applicant's credit history 0 (delay in past payments) 1 (critical account/other credits elsewhere) 2 (no credits taken/all credits paid back duly) 3 (existing credits paid back duly till now) 4 (all credits at this bank paid back duly)
purpose	Purpose of loan 0 (other) 1 (new car) 2 (used car) 3 (furniture/equipment) 4 (radio/television) 5 (domestic appliances) 6 (repairs) 7 (education) 8 (vacation) 9 (retraining) 10 (business)
amount	Credit amount (result of unknown monotonic transformation)

Table 27: Variables of South German Credit dataset (1/3).

VARIABLES	
savings	Savings account balance (in Deutsche Mark) 1 (unknown/ no savings account) 2 (< 100 DM) 3 ($100 \leq \dots < 500$ DM) 4 ($500 \leq \dots < 1000$ DM) 5 (≥ 1000 DM)
employment_duration	Duration of applicant's current employment 1 (unemployed) 2 (< 1 year) 3 ($1 \leq \dots < 4$ years) 4 ($4 \leq \dots < 7$ years) 5 (≥ 7 years)
installment_rate	Installment amount to disposable income ratio [%] 1 (≥ 35) 2 ($25 \leq \dots < 35$) 3 ($20 \leq \dots < 25$) 4 (< 20)
personal_status_sex	Joint encoding of sex and marital status of applicant 1 (male - divorced/separated) 2 (female - non single or male - single) 3 (male - married/widowed) 4 (female - single)
other_debtors	Presence of co-debtor or guarantor 1 (none) 2 (co-applicant) 3 (guarantor)
present_residence	Years living at current address 1 (< 1 year) 2 ($1 \leq \dots < 4$ years) 3 ($4 \leq \dots < 7$ years) 4 (≥ 7 years)
property	Applicant's most valuable property 1 (unknown / no property) 2 (car or other) 3 (building soc. savings agr / life insurance) 4 (real estate)

Table 28: Variables of South German Credit dataset (2/3).

VARIABLES	
age	Applicant's age (years)
other_installment_plans	Installment plans with other banks 1 (bank) 2 (stores) 3 (none)
housing	Type of housing 1 (for free) 2 (rent) 3 (own)
number_credits	Number of credits (ongoing or past, including current) with this bank 1 (1) 2 (2-3) 3 (4-5) 4 (≥ 6)
job	Applicant's job and employability 1 (unemployed/ unskilled - non-resident) 2 (unskilled - resident) 3 (skilled employee / official) 4 (manager / self-empl. / highly qualif. employee)
people_liable	Number of people who financially depend on the applicant 1 (3 or more) 2 (0 to 2)
telephone	Presence of telephone landline registered under applicant's name (2) or not (1)
foreign_worker	Foreign worker (1) or not (2)
credit_risk	Punctually payed back every installment (1) or not (2)

Table 29: Variables of South German Credit dataset (3/3).

D.2.4 Statistics

STATISTICS						
Ordinal						
name	type	count	unique	mostFrequent	leastFrequent	missing
status	string	1000	4	4 (≥ 200)	3 ($0 \leq \dots < 200$)	0
savings	string	1000	5	1 (unknown/no savings)	4 ($500 \leq \dots < 1000$)	0
employment_duration	string	1000	5	3 ($1 \leq \dots < 4$)	1 (unemployed)	0
installment_rate	string	1000	4	4 (< 20)	1 ≥ 35	0
present_residence	string	1000	4	4 (≥ 7 yrs)	1 (< 1 yr)	0
number_credits	string	1000	4	1 (1)	4 (≥ 6)	0
people liable	string	1000	2	2 (0 to 2)	1 (3 or more)	0

Table 30: Ordinal variables statistics of South German Credit dataset

Categorical						
name	type	count	uniqueEntries	mostFrequent	leastFrequent	missing
credit_history	string	1000	5	2 (no credits taken)	0 (delay in paying off)	0
purpose	string	1000	11	3 (furniture/equipment)	8 (vacation)	0
status_sex	string	1000	4	3 (male-marr/widow)	1 (male-divorc/separ)	0
other_debtors	string	1000	3	1 (none)	2 (co-appliant)	0
property	string	1000	4	3 (building soc. savings)	4 (real estate)	0
other_plans	string	1000	3	3 (none)	2 (stores)	0
housing	string	1000	3	2 (rent)	3 (own)	0
job	string	1000	4	3 (skilled empl/office)	1 (unempl/unsk non-res)	0
telephone	string	1000	2	1 (no)	2 (yes)	0
foreign_worker	string	1000	2	2 (no)	1 (yes)	0
credit_risk	string	1000	2	1 (good)	0 (bad)	0

Table 31: Categorical variables statistics of South German Credit dataset

Quantitative									
name	type	count	min	median	max	mean	stdDev	miss	zeros
duration	number	1000	4	18	72	20.90	12.06	0	0
amount	number	1000	250	2319.50	18424	3271.25	2822.75	0	0
age	number	1000	19	33	75	35.54	11.35	0	0

Table 32: Quantitative variables statistics of South German Credit dataset.